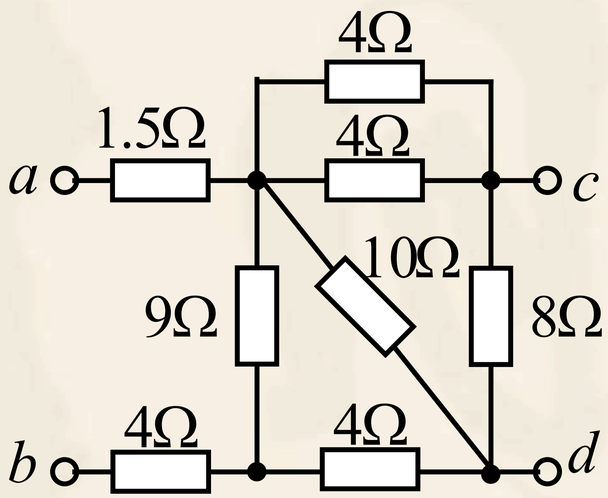
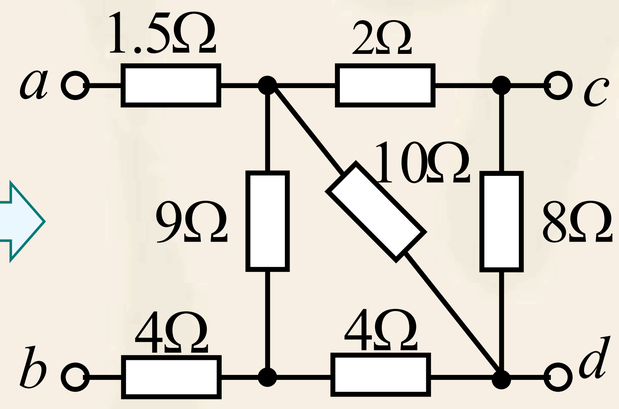
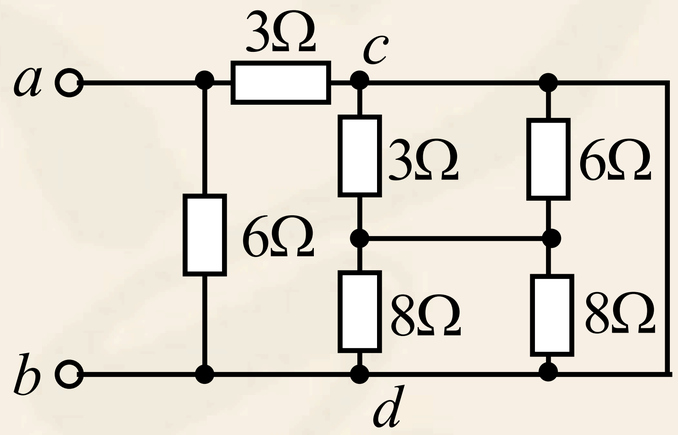
求图示电路的等效电阻 R_{ab}, R_{cd}



（1）





（2）

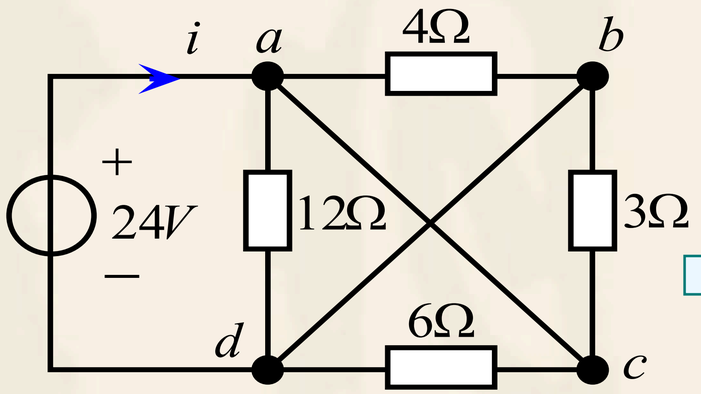
解： 通过同一电流的元件为串联；两端为同一电压的元件为并联；无电流通过的元件可开路；电位相同的节点可短路。因此

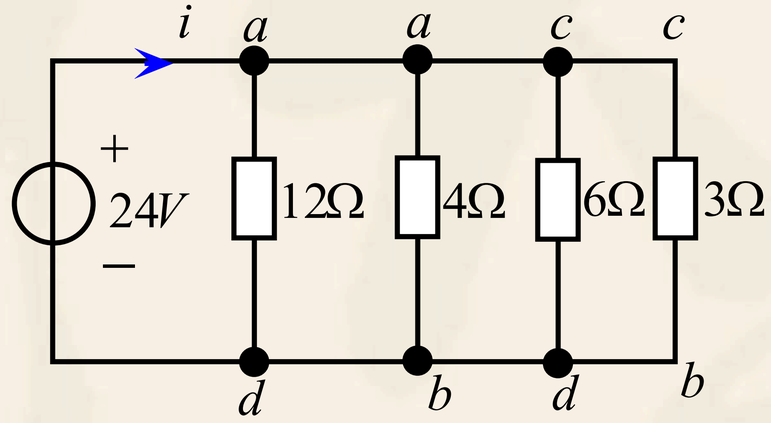
（1）$R_{ab} = 1.5 + 9 // [(2+8) // 10 + 4] + 4 = 1.5 + 4.5 + 4 = 10\Omega$

$R_{cd} = 8 // [2 + (9+4) // 10] = 3.9\Omega$

（2）$R_{ab} = 3 // 6 = 2\Omega$　　$R_{cd} = 0\Omega$

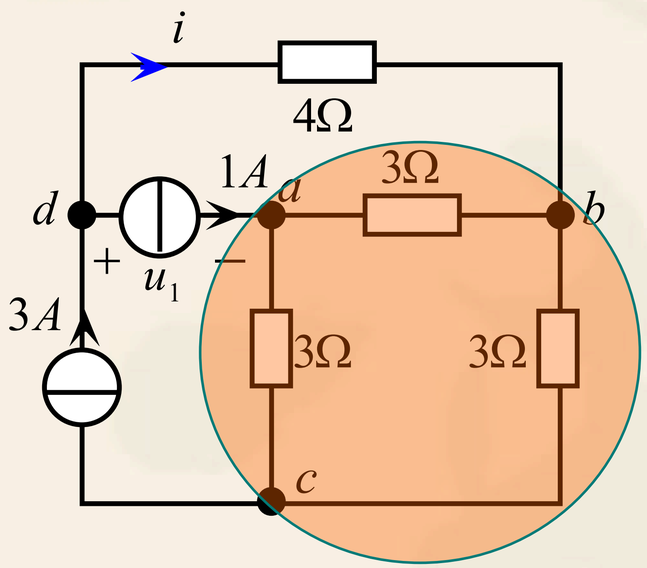
图示电路，求电流i.

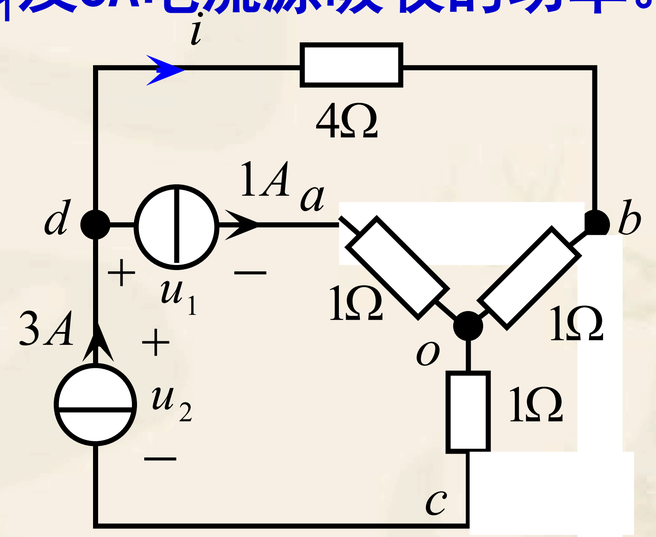




$$i = 20A$$

例5 图示电路，求电压u_1及3A电流源吸收的功率。





$$i = 2A$$

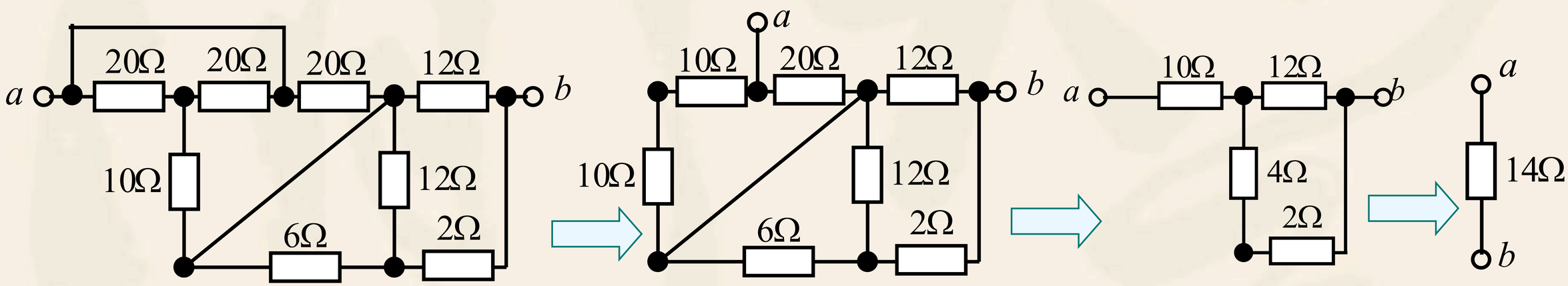
$$u_{bc} = 1 \times i + 1 \times 3 = 5V$$

$$\boldsymbol{u_2 = i \times 4 + u_{bc} = 13V}$$

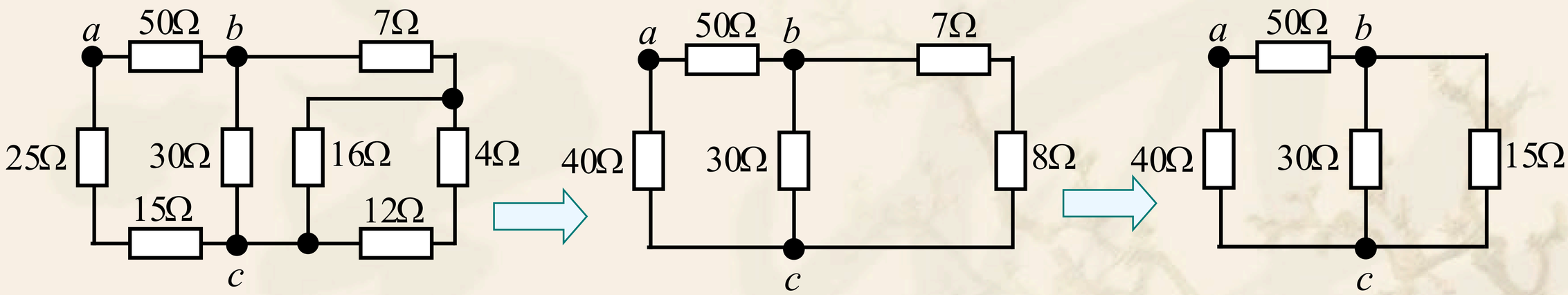
$$P_{3A} = -3u_2 = -39W$$

$$u_1 = u_2 - u_{oc} - u_{ao} = 13 - 1 \times 3 - 1 \times 1 = 9V$$

求图示单口电路的等效电路。



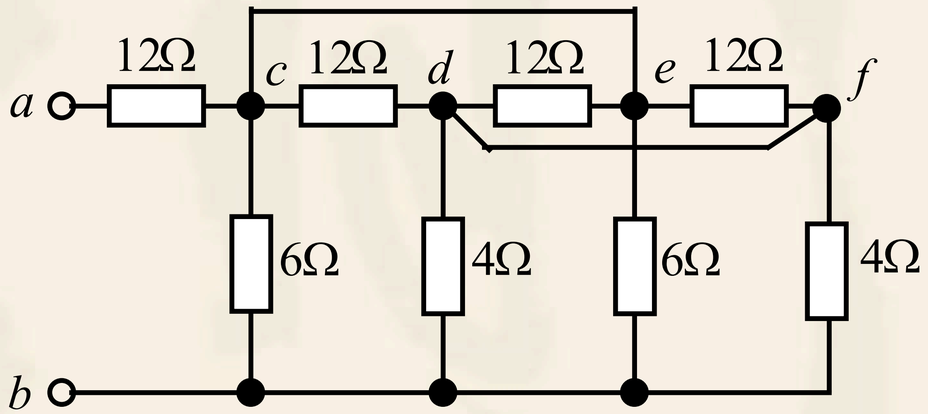
求图示等效电阻 R_{ab}, R_{bc}, R_{ca}

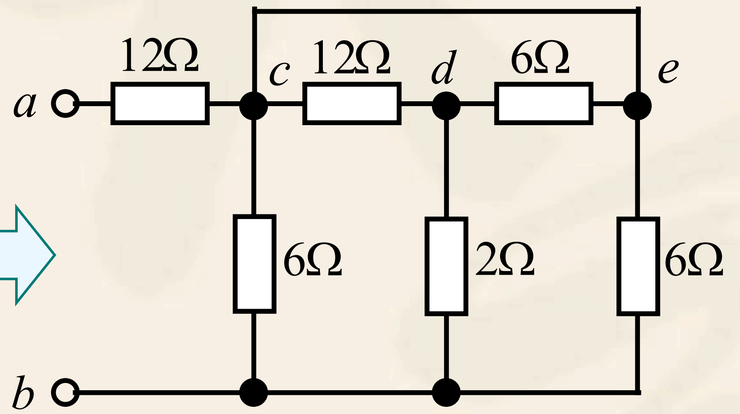


$$R_{ab} = 50 // [40 + (30 // 15)] = 25\Omega \qquad R_{bc} = (40 + 50) // 30 // 15 = 90 // 10 = 9\Omega$$

$$R_{ac} = 40 // [50 + (30 // 15)] = 40 // 600 = 24\Omega$$

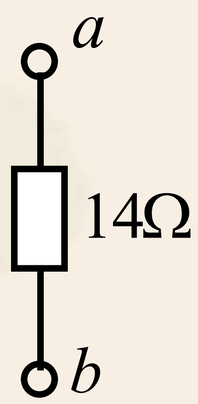
求图示单口电路的等效电阻 R_{ab}



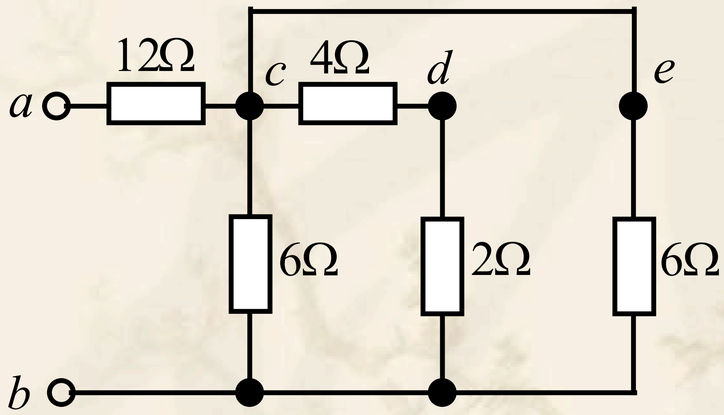


解：　等效变换

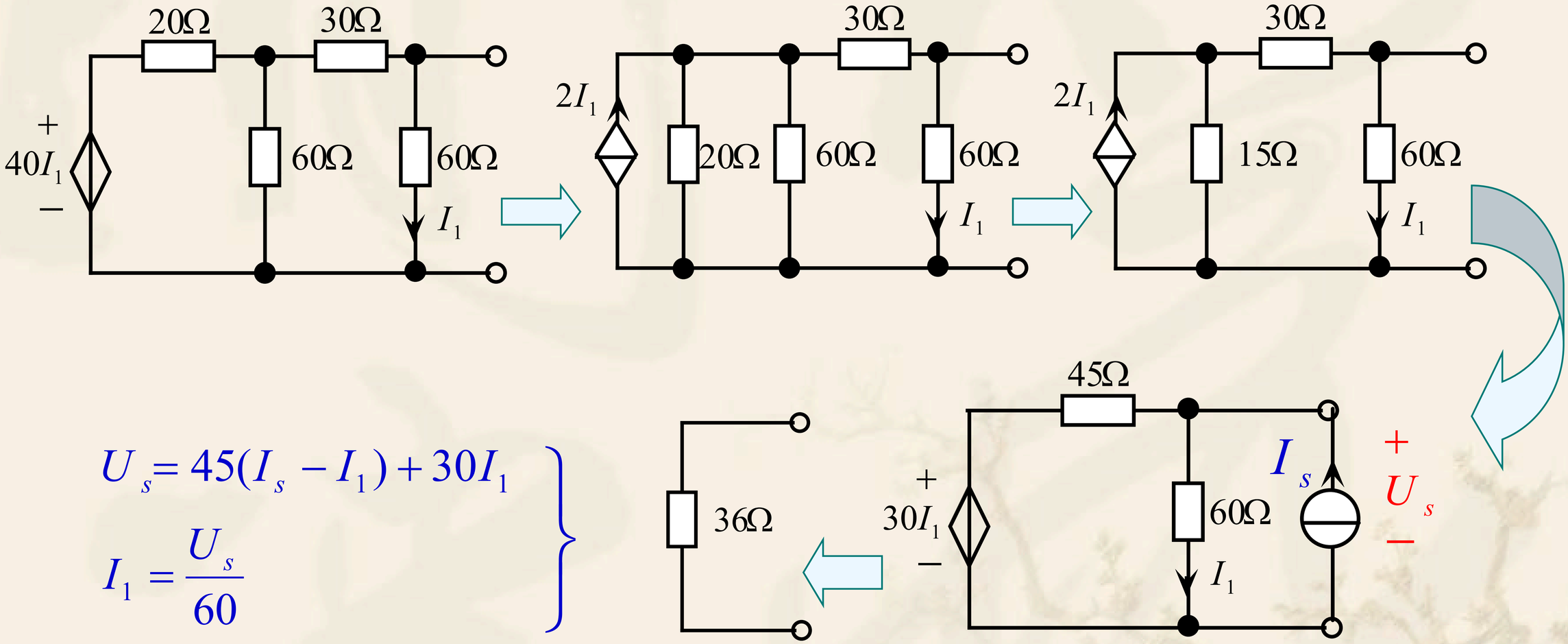
$$R_{ab} = 14\Omega$$



等效电路



例7 求图示单口电路的输入电阻。



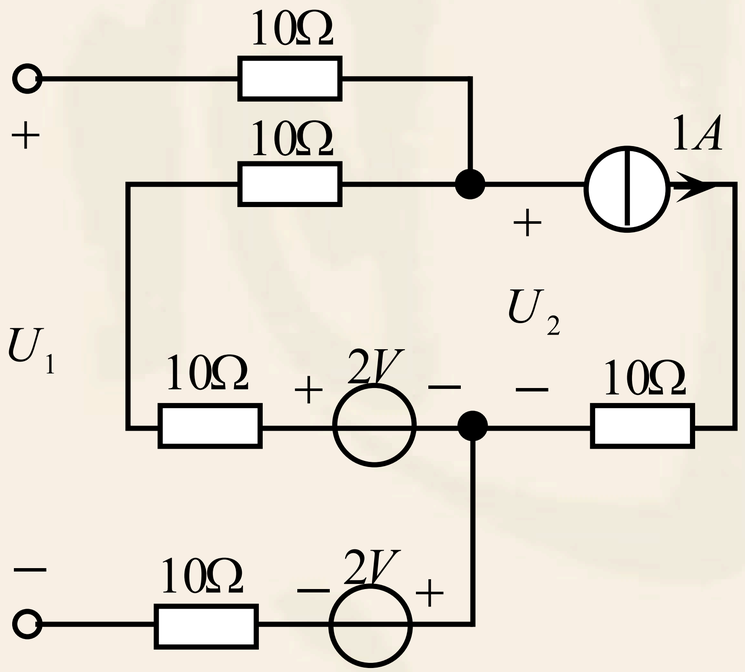
$$\left.\begin{aligned} U_s &= 45(I_s - I_1) + 30I_1 \\ I_1 &= \frac{U_s}{60} \end{aligned}\right\}$$

$$R_0 = \frac{U_s}{I_s} = 36\Omega$$

主观题 10分

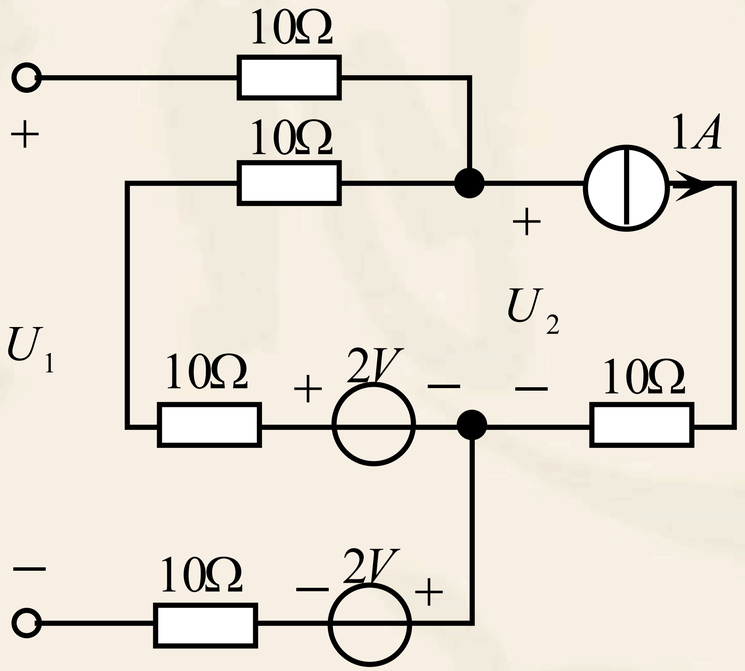
设置

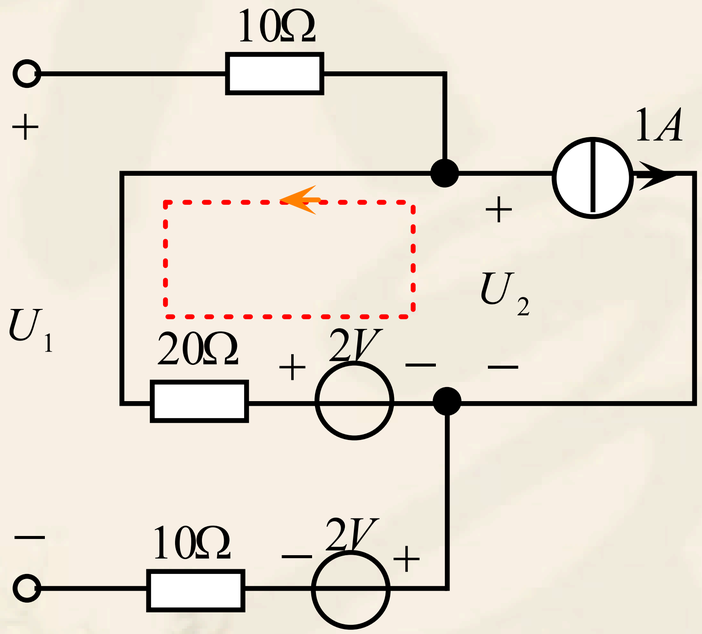
求U_1，U_2，及1A电流源吸收的功率P。



作答

例2 求U_1，U_2，及1A电流源吸收的功率P。





$$P_{1A} = (U_2 - 10\times 1)\times 1 = -28V$$

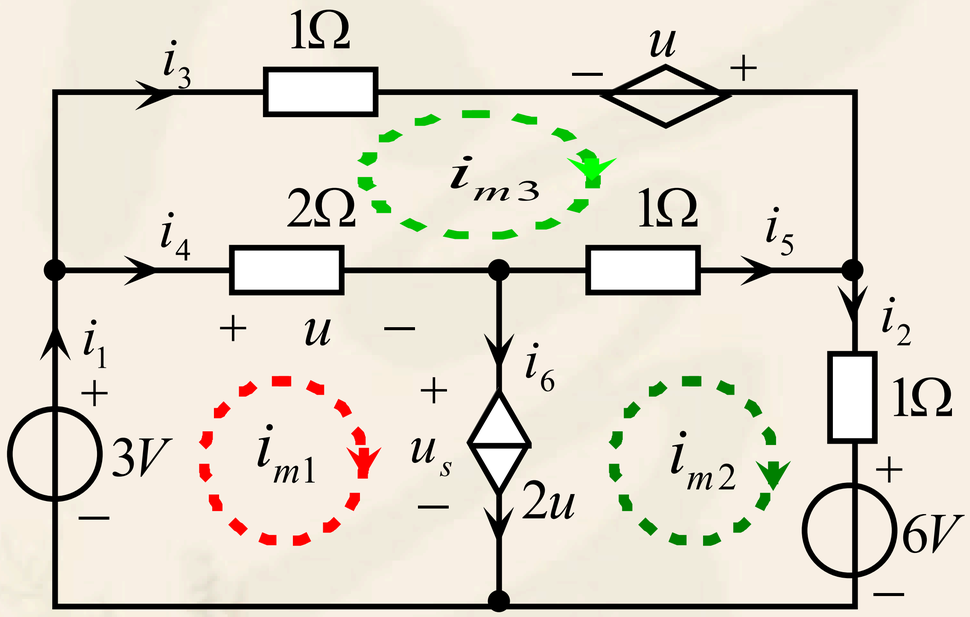
$$U_2 = -20\times 1 + 2 = -18V$$

$$U_1 = U_2 + 2 = -16V$$

图示电路，用网孔法电流法求各支路电流，并求受控源发出的功率。

解：

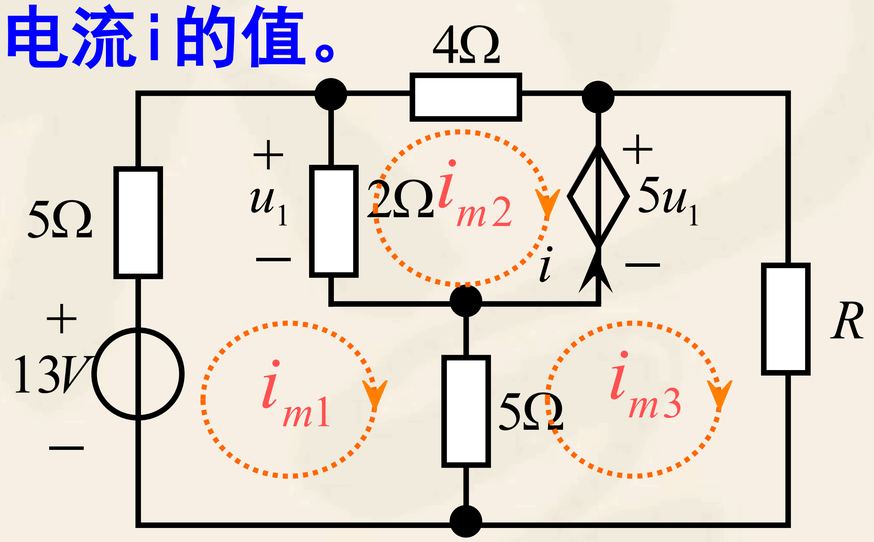
$$\begin{cases} 2i_{m1}-2i_{m3}=3-u_s \\ 2i_{m2}-i_{m3}=u_s-6 \\ -2i_{m1}-i_{m2}+4i_{m3}=u \\ i_{m1}-i_{m2}=2u \\ u=2(i_{m1}-i_{m3}) \end{cases}$$



$$i_{m1}=2A, i_{m2}=-2A, i_{m3}=1A, u=2V, u_s=1V$$

$$\begin{cases} i_1=2A \\ i_2=-2A \\ i_3=1A \\ i_4=1A \\ i_5=-3A \\ i_6=4A \end{cases} \qquad P=-u_s i_6=4W$$

图示电路，$u_1 = 2V$ ，网孔法求电阻R和电流i的值。



$$(5+2+5)i_{m1} - 2i_{m2} - 5i_{m3} = 13$$

$$-2i_{m1} + 6i_{m2} = -5u_1 = -10$$

$$-5i_{m1} + (5+R)i_{m3} = 5u_1 = 10$$

$$u_1 = 2(i_{m1} - i_{m2}) = 2$$

$$i_{m1} = -1A$$

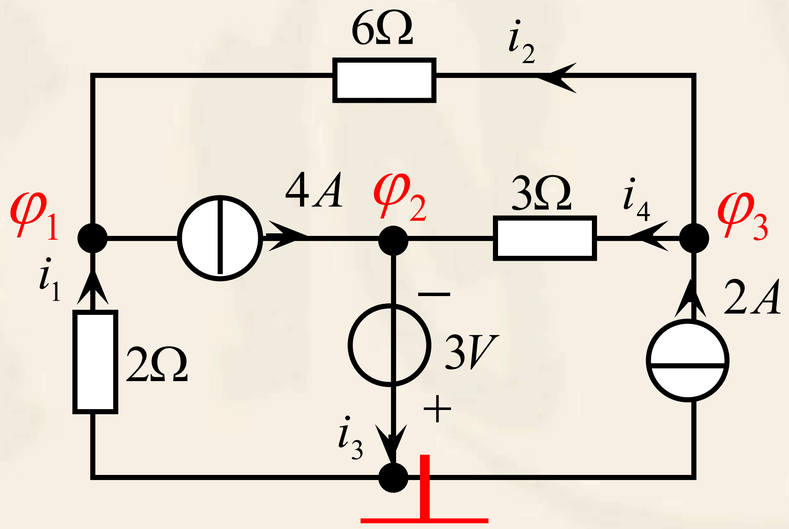
$$i_{m2} = -2A$$

$$i_{m3} = 1A$$

$$R = 0$$

$$i = i_{m3} - i_{m2} = 3A$$

图示电路，用节点法求支路电流 i_1，i_2，i_3，i_4。



节点电位法

$$\left.\begin{aligned}&(\frac{1}{2}+\frac{1}{6})\phi_1-\frac{1}{6}\phi_3=-4\\&\varphi_2=-3V\\&-\frac{1}{6}\varphi_1--\frac{1}{3}\varphi_2+(\frac{1}{3}+\frac{1}{6})\varphi_3=2\end{aligned}\right\}$$

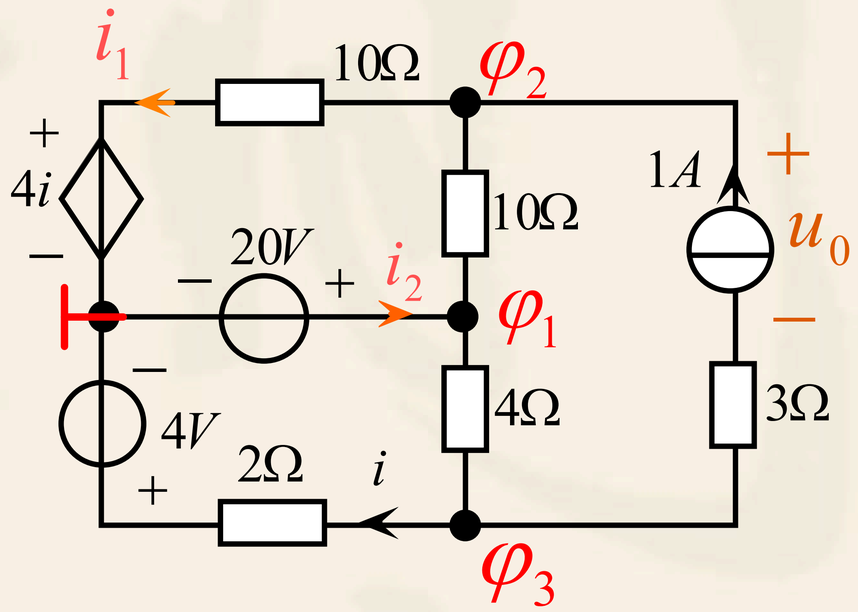
$$i_1=-\frac{\varphi_1}{2},i_2=\frac{\varphi_3-\varphi_1}{6}$$

$$i_4=\frac{\varphi_3-\varphi_2}{3}$$

$$i_3=i_4+4$$

$$i_1=3A,i_2=1A,i_3=5A,i_4=1A$$

用节点法求图示电路中各个独立源发出的功率及受控源吸收的功率。



$$\left.\begin{aligned}
&\varphi_1 = 20V \\
&-\frac{1}{10}\varphi_1 + (\frac{1}{10}+\frac{1}{10})\varphi_2 = 1+\frac{4i}{10} \\
&-\frac{1}{4}\varphi_1 + (\frac{1}{2}+\frac{1}{4})\varphi_3 = -1+\frac{4}{2} \\
&\varphi_3 = 2i+4
\end{aligned}\right\}$$

$$\left.\begin{aligned}
&\varphi_1 = 20V \\
&\varphi_2 = 19V \\
&\varphi_3 = 8V \\
&i = 2A
\end{aligned}\right\}$$

$$i_1 = \frac{\varphi_2 - 4i}{10} = 1.1A$$

$$i_2 = i + i_1 = 3.1A$$

$$u_0 = \varphi_2 - \varphi_3 + 3\times 1 = 14V$$

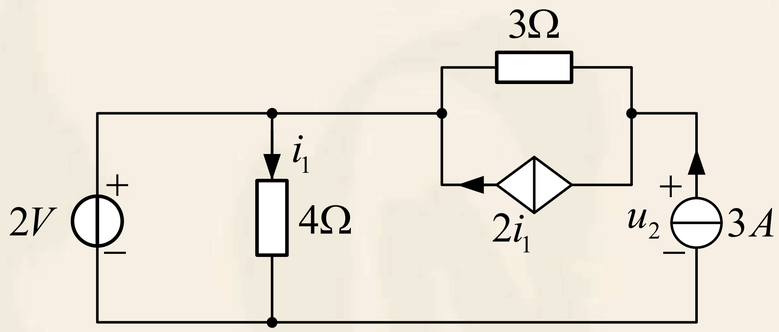
$$P_{4V} = -4i = -8W$$

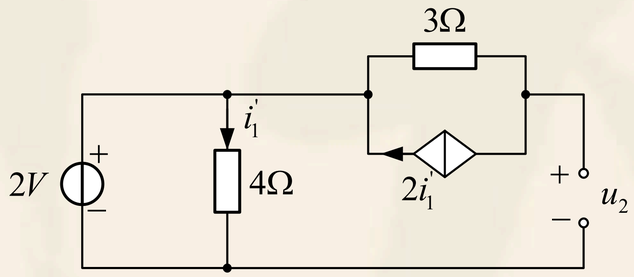
$$P_{20V} = 20i_2 = 62W$$

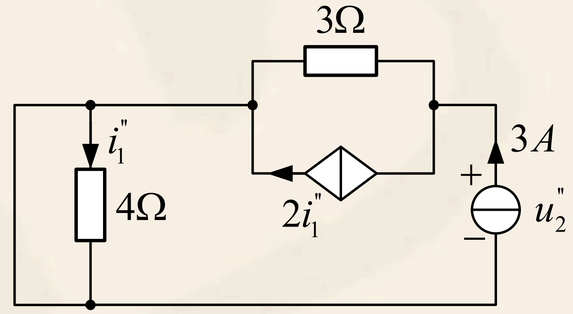
$$P_{1A} = 1\times u_0 = 14W$$

$$\boldsymbol{P_{4I} = 4i\times i_1 = 8.8W}$$

如图所示电路，用叠加定理求电压u_2







2V电压单独作用时的分电路如图（a）所示，

$$i_1' = \frac{2}{4}A = 0.5A$$

故　　$u_2' = -3 \times 2i_1' + 2 = -1V$

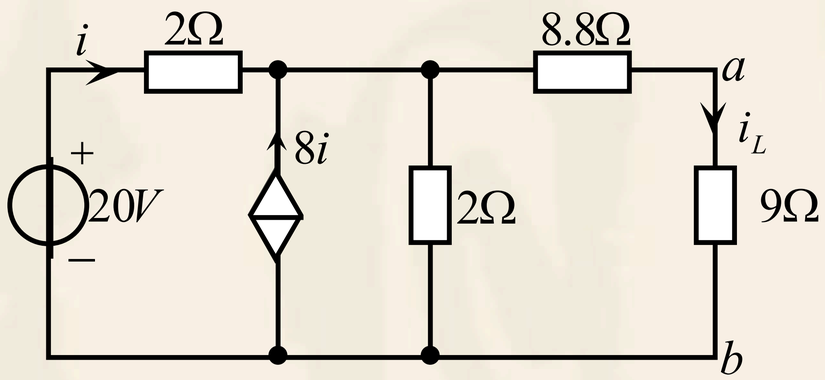
$3A$电流源单独作用的分电路如图（b）所示，

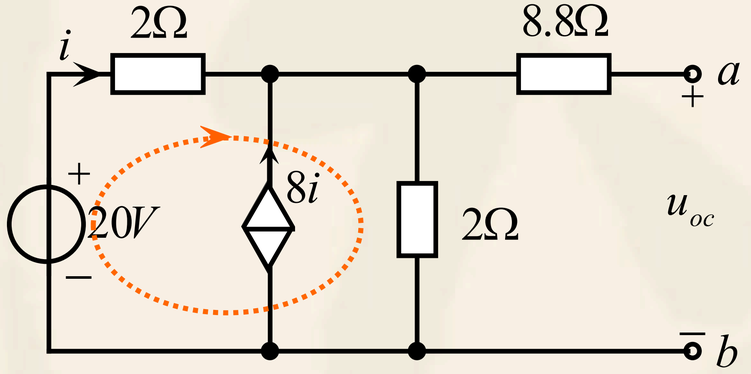
$$i_1'' = 0 \qquad u_2'' = 3 \times 3V = 9V$$

由叠加定理得到。

$$u_2 = u_2' + u_2'' = (-1+9)V = 8V$$

用戴维南定理求图示电路中的电流 i_L 。



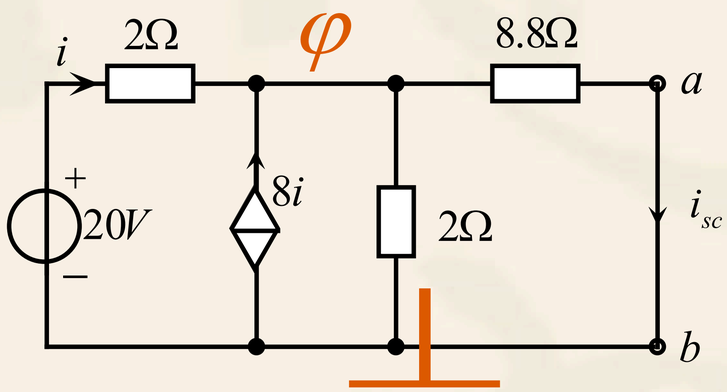


$$2i + 2(i + 8i) = 20$$

$$i = 1A$$

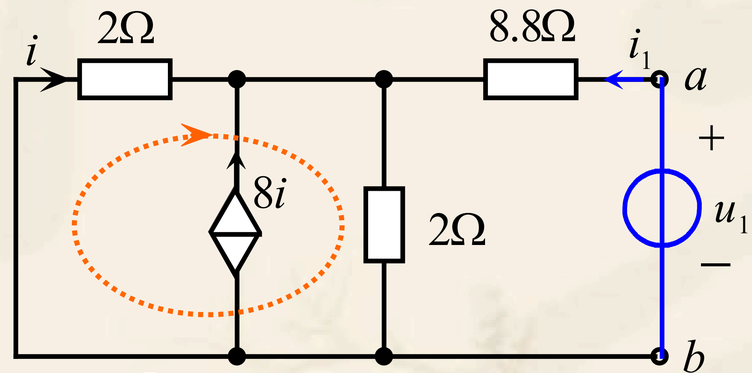
$$u_{oc} = 2(i + 8i) = 18V$$

解：



$$\left.\begin{aligned}(\frac{1}{2}+\frac{1}{2}+\frac{1}{8.8})\varphi = \frac{20}{2}+8i \\ i = \frac{20-\varphi}{2}\end{aligned}\right\}$$

$$\varphi = 17.6V \qquad i_{sc} = \frac{\varphi}{8.8} = 2A$$

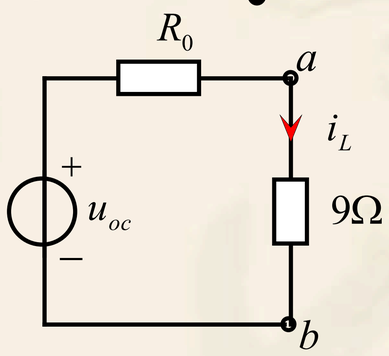


$$2i + 2(9i + i_1) = 0$$

$$i = -0.1i_1$$

$$u_1 = 8.8i_1 - 2i = 9i_1$$

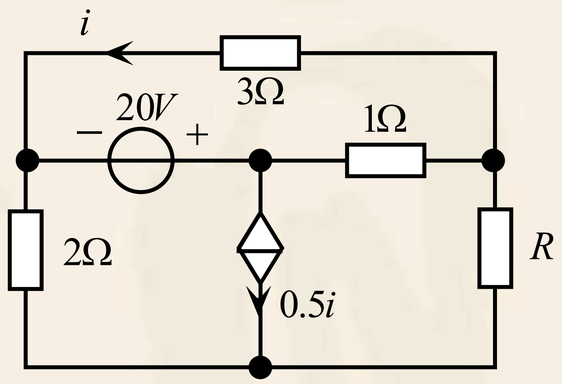
$$R_0 = \frac{u_1}{i_1} = 9\Omega$$

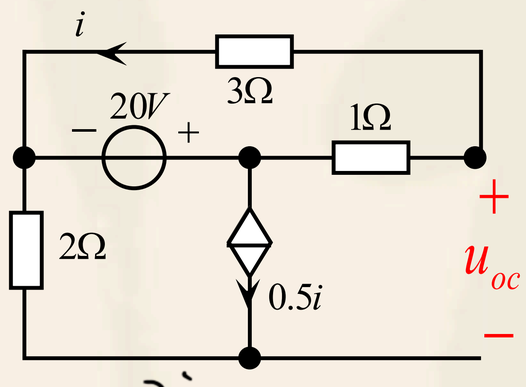


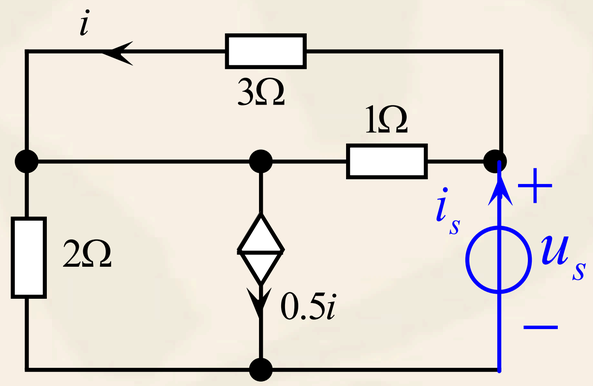
$$i_L = \frac{u_{oc}}{R_0 + 9} = 1A$$

$$R_0 = \frac{u_{oc}}{i_{sc}} = 9\Omega$$

例3　图示电路，R为多大时可获得最大功率并求该最大功率。







（1）求开路电压　$4i = 20, i = 5A$　$u_{oc} = 3i + 2(-0.5i) = 2i = 10V$

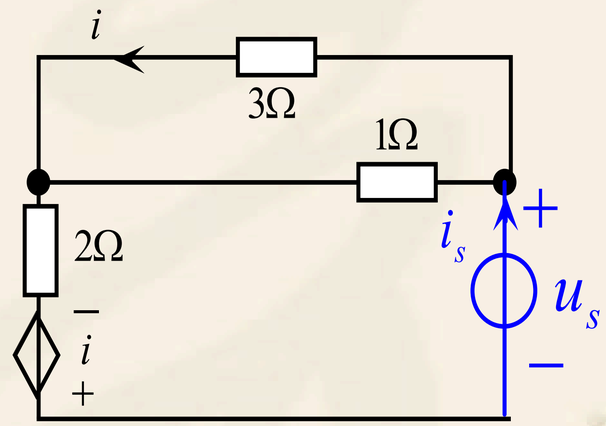
（2）求等效电阻　$u_s = 3i + 2i_s - i = 2i + 2i_s$

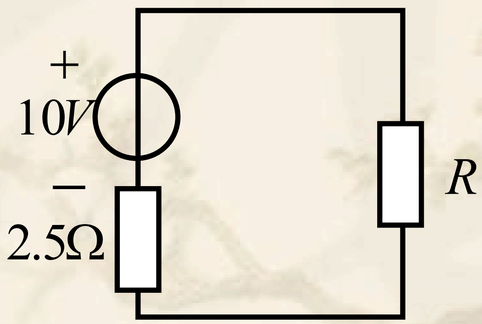
$$i = \frac{1}{4} i_s \qquad R_0 = \frac{u_s}{i_s} = 2.5\Omega$$

（3）画出戴维南等效电路

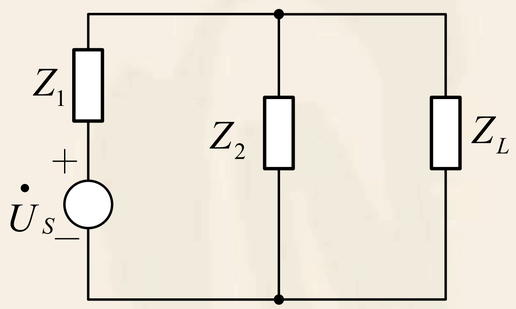
（4）应用最大功率传输定理

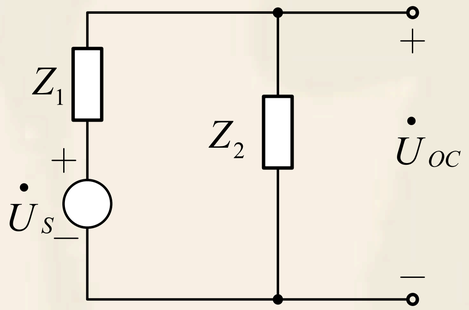
$$R = R_0 = 2.5\Omega, \qquad P_m = \frac{u_{oc}^2}{4R_0} = 10W$$

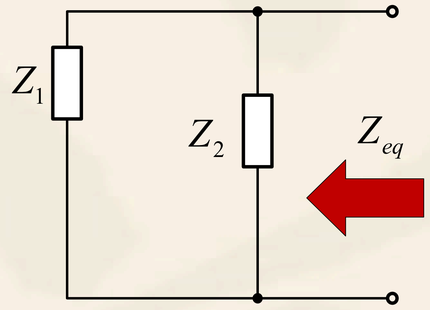




如图所示的正弦稳态电路：$\dot{U}_S = 9\angle 0^\circ$ ，$Z_1 = -j2\Omega$ ，$Z_2 = (2+j2)\Omega$。当负载阻抗Z为何值时可获得最大功率，并求Pmax。







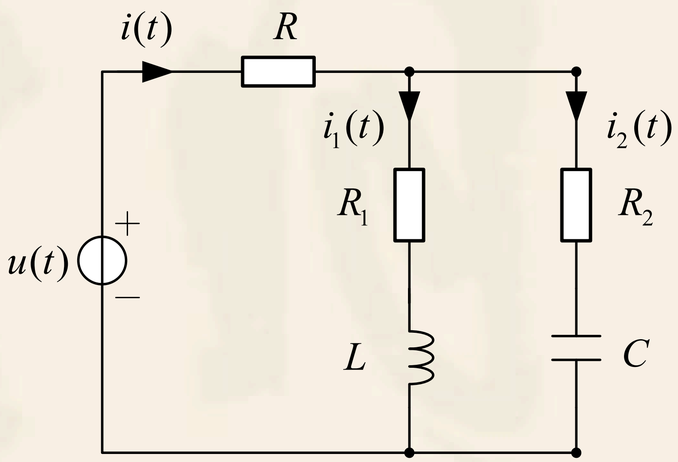
$$\dot{U}_{OC} = \dot{U}_S \times \frac{Z_2}{Z_1 + Z_2} = 9\sqrt{2}\angle 45°V$$

$$Z_{eq} = \frac{Z_1 \times Z_2}{Z_1 + Z_2} = (2 - j2)\Omega$$

$$Z = Z_0^* = (2 + j2)\Omega$$

$$P_{\max} = \frac{U_{oc}^2}{4\mathrm{R}_{eq}} = 20.25W$$

如图所示电路，已知 $u(t)=50+100\sqrt{2}\cos 5000tV$，$R=10\Omega$，$R_1=10\Omega$，$R_2=10\Omega$，$L=2mH$，$C=20\mu F$，**求** $i(t)$，$i_1(t)$，$i_2(t)$。



（1）直流分量单独作用时，电感短路，电容短路

$$I_0=\frac{50}{10+10}=2.5A$$

$$I_{10}=I_0$$

$$I_{20}=0$$

（2） $u_1(t)=100\sqrt{2}\cos 5000tV$ **单独作用时，电感短路，电容短路**

$$\dot{U}_1=100\angle 0^\circ,\ j\omega_1 L=j10\Omega,\ \frac{1}{j\omega_1 C}=-j10\Omega,\ Z_1=10+j10\Omega,\ Z_2=10-j10\Omega$$

$$Z_{12}=\frac{Z_1Z_2}{Z_1+Z_2}=10\Omega$$

$$\dot{I}_1=\frac{\dot{U}_1}{20}=5\angle 0^\circ A$$

$$\dot{I}_{11}=\frac{Z_2}{Z_1+Z_2}\dot{I}_1=\frac{5}{\sqrt{2}}\angle -45^\circ A$$

$$\dot{I}_{21}=\frac{Z_1}{Z_1+Z_2}\dot{I}=\frac{5}{\sqrt{2}}\angle 45^\circ A$$

$$i(t)=2.5+5\sqrt{2}\cos(5000t)A$$

$$i_1(t)=2.5+5\cos(5000t-45^\circ)A$$

$$i_2(t)=5\cos(5000t+45^\circ)A$$